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DEVICE

Abstract

The device includes: a identifying unit that specifies a corresponding moving body to which a defective component evacuated from a component line through which a plurality of components flow is assembled among one or more moving bodies moving the manufacturing line by unmanned operation, wherein the component line merges with the manufacturing line in an assembly area for performing assembly of the component to the moving body; an evacuation instruction for evacuating the corresponding moving body from the manufacturing line by unmanned operation; and an instruction unit that issues at least one of a skip instruction for skipping assembly of the corresponding moving body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims priority to Japanese Patent Application No. 2024-028020 filed on Feb. 28, 2024, incorporated herein by reference in its entirety.
BACKGROUND
1. Technical Field
[0002] The present disclosure relates to a device.
2. Description of Related Art
[0003] Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2017-538619 (JP 2017-538619 A) discloses technology for causing a vehicle to travel by unmanned driving in a manufacturing process of the vehicle.
SUMMARY
[0004] In an assembly area for assembling components to moving bodies such as vehicles, a technology is known in which the moving bodies and the components are merged, and the moving bodies and the components that are merged are assembled. Now, it is conceivable to transport the moving body to the assembly area by unmanned driving. However, appropriately executing assembly, when a component to be assembled to a moving body that is transported by unmanned driving has a defect, has not been studied.
[0005] The present disclosure can be realized as the following aspects. [0006] (1) According to a first aspect of the present disclosure, a device is provided. This device includes an identifying unit that identifies, out of one or more moving bodies that move over a manufacturing line by unmanned driving, a relevant moving body to which a defective component that was evacuated from a component line, over which a plurality of components flows, is to be assembled, in which the component line merges with the manufacturing line in an assembly area for executing assembly of the component to the moving body, and an instruction unit for issuing at least one of an evacuation instruction for evacuating the relevant moving body from the manufacturing line by the unmanned driving, and a skip instruction for skipping the assembling to the relevant moving body. According to this aspect, an incorrect component can be suppressed from being assembled to the moving body in the assembly area, and assembly in the assembly area can be appropriately executed. [0007] (2) In the above aspect, when the evacuation instruction is issued, the instruction unit may issue a first entry instruction for instructing equipment that is configured to change a position of each of the components flowing over the component line, to cause to enter, into a first vacant region generated by the defective component being removed from the component line, the component trailing the defective component, and a second entry instruction for instructing a trailing moving body that is the moving body trailing the relevant moving body, to enter a second vacant region generated by the relevant moving body being removed from the manufacturing line. According to this aspect, each component can be efficiently moved on the component line, and also each moving body can be efficiently moved on the manufacturing line, and accordingly, assembly can be executed more efficiently in the assembly area. [0008] (3) In the above aspect, the instruction unit may cause the trailing moving body to enter the second vacant region by making a degree of deceleration of the trailing moving body to be smaller than a degree of deceleration of the moving body preceding the trailing moving body, in the second entry instruction. According to this aspect, a work process executed upstream of the assembly area in the manufacturing line can be suppressed from being affected by speed of the moving body being high, in comparison with when

entering the moving body to the vacant region by further accelerating the trailing moving body. [0009] (4) In the above aspect, the instruction unit may issue an instruction to repair a defect of the defective component that was evacuated, and when repair of the defect is complete, issue an instruction for causing equipment that is configured to enter a repaired component that is the component regarding which the repair is completed to the component line, to enter the repaired component to the component line, and an instruction for causing the relevant moving body evacuated by the evacuation instruction to enter the manufacturing line by the unmanned driving. According to this aspect, the repaired component can be returned to the component line and the relevant moving body can be returned to the manufacturing line, without manual work. [0010] (5) In the above aspect, the instruction unit may issue an instruction to repair a defect of the defective component that was evacuated, and when repair of the defect is complete, issue an instruction for moving a repaired component that is the component regarding which the repair is completed, toward the relevant moving body using a device different from the component line, and an instruction for assembling the repaired component to the relevant moving body. According to this aspect, the repaired component can be assembled to the relevant moving body without entering the repaired component to the component line again. The present disclosure can also be realized in a form of, for example, a moving body, a system, a method of manufacturing the moving body, a method of controlling the moving body, a program, a non-transitory recording medium in which the program is recorded, a program product, and so forth, in addition to the form of a device that is described above. Note that the program product may also be provided as a storage medium that stores the program, or may be provided as a program product that can be distributed via a network, for example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0012] FIG. **1** is a conceptual diagram showing a configuration of a system according to a first embodiment;

[0013] FIG. **2** is a conceptual diagram illustrating component evacuation and component return;

[0014] FIG. **3** is a conceptual diagram illustrating a correspondence relationship between each vehicle and each component;

[0015] FIG. **4** is a block diagram illustrating a configuration of a system;

[0016] FIG. **5** is a flowchart illustrating a processing procedure of travel control of a vehicle according to the first embodiment;

[0017] FIG. **6** is a flowchart of a manufacturing process according to the first embodiment;

[0018] FIG. **7** is a diagram illustrating an example of a manufacturing process according to the first embodiment;

[0019] FIG. **8** is a block diagram illustrating a configuration of a system according to a second embodiment;

[0020] FIG. **9** is a flowchart of a manufacturing process according to the second embodiment;

[0021] FIG. **10** is a diagram illustrating an example of a manufacturing process according to the second embodiment;

[0022] FIG. **11** is a block-diagram illustrating a configuration of a system according to a third embodiment; and

[0023] FIG. **12** is a flowchart illustrating a processing procedure of travel control of the vehicle according to the third embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

A. First Embodiment

[0024] FIG. **1** is a conceptual diagram illustrating a configuration of a system **50** according to a first embodiment. The system **50** includes one or more vehicles **100** as moving bodies, a server **200**, and one or more external sensors **300**. The server **200** in the first embodiment corresponds to a "device" in the present disclosure.

[0025] In the present disclosure, "moving body" means a movable object, and is, for example, a vehicle or an electric vertical takeoff and landing machine (a so-called flying vehicle). The vehicle may be a vehicle traveling by a wheel or a vehicle traveling by an infinite track, and is, for example, a passenger car, a truck, a bus, a two-wheeled vehicle, a four-wheeled vehicle, a tank, a construction vehicle, or the like. Vehicles include battery electric vehicle (BEV: Battery Electric Vehicle), gasoline-powered vehicles, hybrid electric vehicle, and fuel cell electric vehicle. When the moving body is other than the vehicle, the expressions of "vehicle" and "vehicle" in the present disclosure can be appropriately replaced with "moving body", and the expression of "traveling" can be appropriately replaced with "moving".

[0026] The vehicle **100** is configured to be able to travel by unmanned driving. The term "unmanned driving" means driving that does not depend on the traveling operation of the passenger. The traveling operation means an operation related to at least one of "running", "turning", and "stopping" of the vehicle **100**. The unmanned driving is realized by automatic or manual remote control using a device located outside the vehicle **100** or by autonomous control of the vehicle **100**. A passenger who does not perform the traveling operation may be on the vehicle **100** traveling by the unmanned driving. The passenger who does not perform the traveling operation includes, for example, a person who is simply seated on the seat of the vehicle **100** and a person who performs a work different from the traveling operation such as an assembling operation, an inspection operation, and an operation of switches while riding on the vehicle **100**. Driving by the traveling operation of the occupant is sometimes referred to as "manned driving".

[0027] Herein, "remote control" includes "full remote control" in which all of the operations of the vehicle **100** are completely determined from the outside of the vehicle **100**, and "partial remote control" in which a part of the operations of the vehicle **100** is determined from the outside of the vehicle **100**. Further, "autonomous control" includes "fully autonomous control" in which the vehicle **100** autonomously controls its operation without receiving any information from a device external to the vehicle **100**, and "partially autonomous control" in which the vehicle **100** autonomously controls its operation using information received from a device external to the vehicle **100**.

[0028] The vehicle **100** may be provided with a configuration that can be moved by unmanned driving, and may be, for example, in the form of a platform having a configuration described below. Specifically, the vehicle **100** may include at least a vehicle control device and an actuator group, which will be described later, in order to perform three functions of "running", "turning", and "stopping" by unmanned driving. When information is acquired from a device outside the vehicle **100** for unmanned driving, the vehicle **100** may further include a communication device. That is, the vehicle **100** that can be moved by the unmanned driving may not be equipped with at least a part of an interior component such as a driver's seat or a dashboard. In addition, the vehicle **100** that can be moved by unmanned driving may not be equipped with at least a part of an exterior component such as a bumper or a fender. In addition, the vehicle **100** that can be moved by unmanned driving may not be equipped with the body shell. In this instance, the remaining components, such as the body shell, may be mounted to the vehicle **100** until the vehicle **100** is shipped from the factory FC. In addition, while the remaining components such as the body shell are not mounted on the vehicle **100**, the remaining components such as the body shell may be mounted on the vehicle **100** after the vehicle **100** is shipped from the factory FC. Each of the components may be mounted from any direction, such as the upper side, lower side, front side, rear side, right side or left side of the vehicle **100**, each may be mounted from the same direction, or may be mounted from a different direction.

[0029] In the present embodiment, the system **50** is used in a factory FC that manufactures the vehicles **100**. The reference coordinate system of the factory FC is a global coordinate system GC, and any position in the factory FC can be represented by the coordinates of X, Y, Z in the global coordinate system GC. Vehicle **100** travels in a factory FC from a first location L**1** to a second location L**2** through a track TR on which vehicle **100** can travel by unmanned driving.

[0030] In the factory FC, a plurality of external sensors **300** are installed along the track TR. The external sensor **300** is a sensor located outside the vehicle **100**. The external sensor **300** is constituted by a camera. The camera as the external sensor **300** captures an image of the vehicle **100** and outputs a captured image as a detection result. The external sensor **300** includes a communication device (not shown), and can communicate with another device such as the server **200** by wired communication or wireless communication. The positions of the external sensors **300** in the factory FC are adjusted in advance.

[0031] The factory FC has a manufacturing line ML and a component line PL. The manufacturing line ML and the component line PL merge in the assembly area AA. The manufacturing line ML is a line for transporting the vehicles **100** by unmanned driving. In the present embodiment, a part of the track TR corresponds to the manufacturing line ML. The component line PL is a line for transporting a plurality of component PT. In the component line PL, each component PT to be assembled to each vehicle **100** to be manufactured in the factory FC flows. In the present embodiment, the conveyor device Cv for conveying the component PT toward the assembly area AA corresponds to the component line PL. The component line PL may be configured to be capable of sequentially conveying the component PT to the assembly area AA. In other embodiments, a line using a hanger that suspends and moves the component PT may be used, or a line using an automatic guided vehicle (AGV) that conveys the component PT may be used. In the assembly area AA, the assembly of the component PT to the vehicles **100** is performed. In the present embodiment, the assembly in the assembly-area AA is performed by the assembly robots **450**. The assembly robots **450** are exemplary devices for assembling component PT to the vehicles **100**.

[0032] In the present embodiment, the component processor PD is disposed in the vicinity of the component line PL. The component processor PD is configured by, for example, a robotic device. The component processor PD is configured to be capable of performing component evacuation and component return described later.

[0033] Vehicle **100** flowing through the manufacturing line ML is capable of vehicle evacuation and vehicle return. “Vehicle evacuation” means that the vehicle **100** is evacuated to the second evacuation location EP**2** outside the manufacturing line ML. The second evacuation location EP**2** is a location for evacuating the vehicles **100** from the manufacturing line ML. “Vehicle return” means that the vehicle **100** returns onto the manufacturing line ML. The second evacuation location EP**2** may be, for example, a track aligned with the manufacturing line ML in the vehicle-width direction of the track TR or a track connected to the manufacturing line ML. In the present embodiment, vehicle evacuation and vehicle return are realized by traveling by the unmanned driving of the vehicle **100**. In other embodiments, vehicle evacuation and vehicle return may be realized by, for example, a traveling operation by an operator, transportation by an operator or a robot using wheels of the vehicle **100**, or transportation without wheels. Since the vehicle **100** is configured to be movable by unmanned driving, vehicle evacuation and vehicle return can be easily performed by traveling of the vehicle **100** or transporting the vehicle **100** on the track TR.

[0034] FIG. **2** is a conceptual diagram illustrating component evacuation and component return. As in the case of the vehicle **100**, the component PT flowing through the component line PL can be evacuated and returned to the component. “Part evacuation” means that the component PT is evacuated to the first evacuation location EP**1** outside the component line PL. The first evacuation location EP**1** is a location for evacuating the component PT from the component line PL. In the present embodiment, a repair device **460**, which will be described later, is disposed in the first evacuation location EP**1**. “Component return” means that the component PT returns onto the component line PL. In other embodiments, component evacuation or component return may be performed, for example, by an operator, rather than, for example, a component processor PD. Further, for example, a conveyor device Cv, a hanger, or a AGV used in the component line PL may be configured as a device capable of performing component evacuation or component return. Specifically, for example, when the component line PL is configured as a line using the conveyor device Cv as in the present embodiment, the component evacuation may be executed by changing the conveyance direction of the conveyor in the vicinity of the first evacuation location EP**1**. Further, a return conveyor may be laid on the first evacuation location EP**1**, and component return may be performed using the return conveyor.

[0035] FIG. **3** is a conceptual diagram illustrating a correspondence relation between each vehicle **100** and each component PT in the component line PL in the manufacturing line ML. In the vehicle **100***a*, **100***b*, **100***c*, **100***d* illustrated in FIG. **3**, component PTa, PTb, PTc, PTd corresponding to the respective vehicles **100** in a one-to-one manner are assembled. The correspondence between each vehicle **100** and each component PT is managed using, for example, the identification number of the vehicle **100**. The vehicles **100** illustrated in FIG. **3** arrive at the assembly area AA in the order of vehicle **100***a*, **100***b*, **100***c*, **100***d*. The component PT illustrated in FIG. **2** arrive at the assembly area AA in the order of the component PTa, PTb, PTc, PTd corresponding to the arrival order of the vehicles **100**.

[0036] FIG. **4** is a block diagram illustrating a configuration of the system **50**. The vehicle **100** includes a vehicle control device **110** for controlling each unit of the vehicle **100**, an actuator group **120** including one or more actuators driven under the control of the vehicle control device **110**, and a communication device **130** for wirelessly communicating with an external device such as the server **200**. The actuator group **120** includes an actuator of a driving device for accelerating the vehicle **100**, an actuator of a steering device for changing a traveling direction of the vehicle **100**, and an actuator of a braking device for decelerating the vehicle **100**.

[0037] The vehicle control device **110** includes a computer including a processor **111**, a memory **112**, an input/output interface **113**, and an internal bus **114**. The processor **111**, the memory **112**, and the input/output interface **113** are bidirectionally communicably connected via an internal bus **114**. An actuator group **120** and a communication device **130** are connected to the input/output interface **113**. The processor **111** executes the program PG**1** stored in the memory **112** to realize various functions including functions as the vehicle control unit **115**.

[0038] The vehicle control unit **115** controls the actuator group **120** to cause the vehicle **100** to travel. The vehicle control unit **115** can cause the vehicle **100** to travel by controlling the actuator group **120** using the travel control signal received from the server **200**. The travel control signal is a control signal for causing the vehicle **100** to travel. In the present embodiment, the travel control signal includes the acceleration and the steering angle of the vehicle **100** as parameters. In other embodiments, the travel control signal may include the speed of the vehicle **100** as a parameter in place of or in addition to the acceleration of the vehicle **100**.

[0039] The server **200** includes a computer including a processor **201**, a memory **202**, an input/output interface **203**, and an internal bus **204**. The processor **201**, the memory **202**, and the input/output interface **203** are bidirectionally communicably connected via an internal bus **204**. A communication device **205** for communicating with various devices external to the server **200** is connected to the input/output interface **203**. The communication device **205** can communicate with the vehicle **100** and the terminal device **380** of the user by wireless communication, and can communicate with the external sensors **300**, the assembly robots **450**, the repair device **460**, and the component processor PD by wired communication or wireless communication. The user means a user of the system **50** or the factory FC, and is, for example, an administrator or an operator of the factory FC. The processor **201** executes the program PG**2** stored in the memory **202** to realize various functions including functions as the first acquisition unit **215**, the first identifying unit **220**, the instruction unit **230**, the second acquisition unit **245**, and the second identifying unit **240**. Note that the first identifying unit **220** is also simply referred to as an identifying unit.

[0040] The first acquisition unit **215** acquires defect information. The defect information is information related to a defect of the component PT flowing through the component line PL. Hereinafter, a component PT having a defect is also referred to as a defective component. For example, the first acquisition unit **215** may acquire the defect information by detecting the defect using the external sensor **300**, or may acquire the defect information input by the user. The defect information may be transmitted to the server **200** via the terminal device **380**, for example. The defect information may be, for example, information indicating that there is a defect, information indicating a portion having a defect in the component PT, or information indicating a type of a defect such as a mechanical defect, an electric defect, or an external defect.

[0041] When the defect data is acquired, the first identifying unit **220** specifies the corresponding vehicles corresponding to the defective components from among the plurality of component PT flowing in the component line PL. The corresponding vehicle is the vehicle **100** to which the defective component is assembled if the defective component does not have a defect. For example, the first identifying unit **220** specifies the corresponding vehicle by specifying the identification number of the corresponding vehicle and the order in the manufacturing line ML based on the identification number of the defective component and the order in the component line PL. In the present embodiment, the first identifying unit **220** specifies the corresponding vehicle in substantially the same manner as described above even when the completion information to be described later is acquired.

[0042] The instruction unit **230** in the present embodiment functions as a remote control unit as appropriate. The remote control unit generates a travel control signal and transmits a travel control signal to the vehicle **100** to cause the vehicle **100** to travel by remote control. The instruction unit **230** can also be said to issue a remote instruction for causing the vehicle **100** to travel by remote control to the vehicle **100**.

[0043] The instruction unit **230** according to the present embodiment executes a component evacuation instruction and a vehicle evacuation instruction. Further, the instruction unit **230** in the present embodiment issues a component entry instruction, a vehicle entry instruction, a repair instruction, a component return instruction, and a vehicle return instruction. The vehicle evacuation instruction is also simply referred to as “evacuation instruction”. The component entry instruction is also referred to as a first entry instruction, and the vehicle entry instruction is also referred to as a second entry instruction.

[0044] The component evacuation instruction is an instruction for evacuating the defective component from the component line PL. In the component evacuation instruction according to the present embodiment, the instruction unit **230** transmits a control command for moving the defective component from the component line PL to the first evacuation location EP**1** to the component processor PD.

[0045] The vehicle evacuation instruction is an instruction for evacuating the corresponding vehicle from the manufacturing line ML by an unmanned operation. In the vehicle evacuation instruction according to the present embodiment, the instruction unit **230** generates a travel control signal for causing the corresponding vehicle to travel to the outside of the manufacturing line ML, and transmits the travel control signal to the corresponding vehicle.

[0046] The component entry instruction is to instruct the device configured to change the position of the component PT flowing through the component line PL to enter the trailing component into the first vacant region. The trailing component is a trailing component PT of the defective component in the component line PL. The first vacant region is an area generated when a defective component is evacuated from the component line PL by a component evacuation instruction. That is, the first vacant region corresponds to an area originally occupied by the defective component on the component line PL. In the component entry instruction according to the present embodiment, the instruction unit **230** transmits a control command for moving the trailing component to the first vacant region to the component processor PD.

[0047] The vehicle entry instruction is to instruct the trailing vehicle to enter the second vacant region. The trailing vehicle is a trailing vehicle of the corresponding vehicle in the manufacturing line ML. The second vacant region is an area generated when the corresponding vehicle deviates from the manufacturing line ML according to the vehicle evacuation instruction. That is, the second vacant region corresponds to the area originally occupied by the corresponding vehicles on the manufacturing line ML. In the vehicle entry instruction according to the present embodiment, the instruction unit **230** generates a travel control signal for causing the trailing vehicle to enter the second vacant region, and transmits the travel control signal to the trailing vehicle. Further, in the present embodiment, the travel control signal generated in the vehicle entry instruction is a travel control signal for making the degree of deceleration of the trailing vehicle smaller than the degree of deceleration of the preceding vehicle. The preceding vehicle is the vehicle **100** preceding the trailing vehicle when the corresponding vehicle is evacuated. Specifically, when the preceding vehicle decelerates, the instruction unit **230** generates a travel control signal for preventing the deceleration of the trailing vehicle as compared with the preceding vehicle, and transmits the travel control signal to the trailing vehicle.

[0048] The repair instruction is an instruction for repairing a defect of a defective component. The repair instruction in the present embodiment is an instruction for causing a defective component to be repaired at a repair place. In the present embodiment, the first evacuation location EP**1** corresponds to a repair location. Further, in the present embodiment, the repair instruction is issued to the repair device **460** for repairing the defect. Specifically, in the repair instruction, the instruction unit **230** transmits a control signal for causing the repair to be executed at the repair place to the repair device **460**. The repair device **460** is constituted by, for example, a robot capable of repairing a defect. In other embodiments, the repair instruction may include, for example, an instruction for moving the defective component to the repair place.

[0049] The component return instruction is an instruction for causing the repaired component to enter the component line PL. The repaired part is a component PT in which the repair of the defect is completed, that is, the original defective component. The component return instruction is issued to a device configured to allow the repaired component to enter the component line PL. In the present embodiment, the component return instruction is issued to the component processor PD. Further, the component return instruction is executed when the repair of the defect of the defective component is completed. Specifically, in the present embodiment, the instruction unit **230** issues a component return instruction when the completion information is acquired. The completion information is information related to completion of repair of the defect of the defective component. As will be described later, the completion information is acquired by the second acquisition unit **245**.

[0050] The vehicle return instruction is an instruction for causing the corresponding vehicle evacuated by the vehicle evacuation instruction to enter the manufacturing line ML by unmanned driving. More specifically, the vehicle return instruction is an instruction for causing the corresponding vehicle to enter the second return position on the manufacturing line ML. The second return position is a position corresponding to the first return position on the component line PL. The first return position is a position at which the repaired component returns to the component line PL according to the component return instruction. As will be described later, the second return position is specified by the second identifying unit **240**. The vehicle return instruction is executed when the repair of the defect of the defective component is completed, similarly to the component return instruction. In the vehicle return instruction according to the present embodiment, the instruction unit **230** generates a travel control signal for causing the corresponding vehicle to travel to the second return position on the manufacturing line ML, and transmits the travel control signal to the corresponding vehicle.

[0051] The second acquisition unit **245** acquires the above-described completion information. The completion information may be transmitted to the server **200** via the terminal device **380**, for example. In this case, for example, completion information is input to the terminal device **380** by the user. In addition, the completion information may be transmitted from the device in charge of repair of the defect, that is, for example, the repair device **460** to the server **200**.

[0052] The second identifying unit **240** specifies the second return position according to the first return position. The second identifying unit **240** specifies the second position so that the corresponding vehicles arrive at the assembly area AA in accordance with the repaired component that returns from the first return position to the component line PL and is directed to the assembly area AA. That is, the second position is specified so that the corresponding vehicle arrives at the assembly area AA in the order and timing in which the assembly of the corresponding vehicle in the assembly area AA is correctly performed. In this way, in response to the repaired component being returned to the component line PL, the corresponding vehicle can be returned to the manufacturing line ML, and the assembly of the corresponding vehicle can be performed more smoothly in the assembly area AA. The first return position may be determined in advance according to, for example, the position of the repair place or the route from the component line PL to the repair place. Further, the second identifying unit **240** may specify the first return position using the external sensor **300**, for example.

[0053] FIG. **5** is a flowchart illustrating a processing procedure of travel control of the vehicle **100** according to the first embodiment. In the process of FIG. **5**, the processor **201** of the server **200** functions as a remote control unit by executing a program PG**2**. The processor **111** of the vehicle **100** functions as the vehicle control unit **115** by executing the program PG**1**.

[0054] In S**1**, the processor **201** of the server **200** acquires the vehicle position information using the detection result outputted from the external sensor **300**. The vehicle position information is position information that is a basis for generating a travel control signal. In the present embodiment, the vehicle position information includes the position and orientation of the vehicle **100** in the global coordinate system GC of the factory FC. Specifically, in S**1**, the processor **201** acquires vehicle-position data using captured images acquired from cameras that are the external sensors **300**.

[0055] Specifically, in S**1**, the processor **201** acquires the position of the vehicle **100** by, for example, detecting the outline of the vehicle **100** from the captured image, calculating the coordinate system of the captured image, that is, the coordinates of the positioning point of the vehicle **100** in the local coordinate system, and converting the calculated coordinates into the coordinates in the global coordinate system GC. The outline of the vehicle **100** included in the captured image can be detected by, for example, inputting the captured image into a detection model DM using artificial intelligence. The detection model DM is prepared in the system **50** or outside the system **50**, for example, and stored in the memory **202** of the server **200** in advance. Examples of the detection model DM include a learned machine learning model that is learned so as to realize one of semantic segmentation and instance segmentation. As the machine learning model, for example, a convolutional neural network (hereinafter referred to as a CNN) learned by supervised learning using a learning dataset can be used. The training data set includes, for example, a plurality of training images including the vehicle **100** and a label indicating which of the regions in the training image indicates the vehicle **100** and the regions other than the vehicle **100**. When CNN is learned, the parameters of CNN are preferably updated by back propagation so as to reduce the error between the output-result and-label due to the detection model DM. Further, the processor **201** can obtain the direction of the vehicle **100** by estimating the direction of the movement vector of the vehicle **100** calculated from the position change of the feature point of the vehicle **100** between the frames of the captured image using, for example, the optical flow method.

[0056] In S**2**, the processor **201** of the servers **200** determines the target location to which the vehicles **100** should be heading next. In the present embodiment, the target position is represented by the coordinates of X, Y, Z in the global coordinate system GC. In the memories **202** of the servers **200**, reference path RR that is a route on which the vehicles **100** should travel is stored in advance. The route is represented by a node indicating a starting point, a node indicating a passing point, a node indicating a destination, and a link connecting the respective nodes. The processor **201** uses the vehicle position information and the reference path RR to determine the target position to which the vehicle **100** is to be directed next. The processor **201** determines the target position on the reference path RR ahead of the current position of the vehicles **100**.
[0057] In S**3**, the processor **201** of the server **200** generates a travel control signal for causing the vehicle **100** to travel toward the determined target position. The processor **201** calculates the traveling speed of the vehicle **100** from the transition of the position of the vehicle **100**, and compares the calculated traveling speed with the target speed. The processor **201** generally determines the acceleration so that the vehicle **100** accelerates when the travel speed is lower than the target speed, and determines the acceleration so that the vehicle **100** decelerates when the travel speed is higher than the target speed. In addition, the processor **201** determines the steering angle and the acceleration so that the vehicle **100** does not deviate from the reference path RR when the vehicle **100** is located on the reference path RR. Also, if the vehicle **100** is not located on the reference path RR, in other words, if the vehicle **100** deviates from the reference path RR, the processor **201** determines the steering angle and the acceleration so that the vehicle **100** returns to the reference path RR.
[0058] In S**4**, the processor **201** of the servers **200** transmits the generated travel control signal to the vehicles **100**. The processor **201** repeats acquisition of vehicle position information, determination of a target position, generation of a travel control signal, transmission of a travel control signal, and the like at predetermined intervals.
[0059] In S**5**, the processor **111** of the vehicle **100** receives the travel control signal transmitted from the server **200**. In S**6**, the processor **111** of the vehicle **100** controls the actuator group **120** using the received travel control signal, thereby causing the vehicle **100** to travel at the acceleration and the steering angle represented by the travel control signal. The processor **111** repeatedly receives the travel control signal and controls the actuator group **120** at a predetermined cycle. According to the system **50** of the present embodiment, the vehicle **100** can be driven by remote control, and the vehicle **100** can be moved without using a conveyance facility such as a crane or a conveyor.
[0060] FIG. **6** is a flowchart of a manufacturing process for realizing the manufacturing method of the vehicle **100** according to the present embodiment. The processor **201** executes a manufacturing process at predetermined time intervals, for example.
[0061] In S**105**, the instruction unit **230** determines whether or not the first acquisition unit **215** has acquired the defect data. When the defect information is acquired in S**105**, at S**110**, the instruction unit **230** identifies the defective component based on the defect information, and issues a component evacuation instruction for the identified defective component to the component processor PD. The component processor PD to which the component evacuation instruction has been issued evacuates the defective component to the first evacuation location EP**1**. In S**111**, the instruction unit **230** issues a component entry instruction to the component processor PD. The component processor PD that has issued the component entry instruction causes the trailing component to enter the first vacant region R**1**. In S**112**, the instruction unit **230** issues a repair instruction to the repair device **460**. The repair device **460** to which the repair instruction is issued repairs the defect of the defective component.
[0062] In S**115**, the first identifying unit **220** specifies the corresponding vehicles corresponding to the defective components to be evacuated by S**110**. In S**120**, the instruction unit **230** issues a vehicle evacuation instruction to the corresponding vehicle specified by S**115**. The vehicle control

unit **115** of the corresponding vehicle travels to the second evacuation location EP**2** by using the travel control signal transmitted by S**120**. In S**121**, the instruction unit **230** issues a vehicle entry instruction to the trailing vehicle. The vehicle control unit **115** of the trailing vehicle travels to the second vacant region R**2** by using the travel control signal transmitted by S**121**.

[0063] In S**125**, the instruction unit **230** determines whether or not the completion information has been acquired by the second acquisition unit **245**. When the completion information is acquired by S**125**, at S**130**, the instruction unit **230** identifies the repaired component based on the completion information, and issues a component restoration instruction for the identified repaired component to the component processor PD. The component processor PD that has issued the component return instruction returns the defective component from the first evacuation location EP**1** to the component line PL. In S**135**, the first identifying unit **220** identifies the corresponding vehicles corresponding to the repaired components to be restored in S**130**. In S**140**, the instruction unit **230** issues a vehicle return instruction to the corresponding vehicle specified by S**135**. The vehicle control unit **115** of the corresponding vehicle travels from the second evacuation location EP**2** to the manufacturing line ML by using the travel control signal transmitted by S**140**.

[0064] FIG. **7** is a diagram illustrating an example of a manufacturing process. FIG. **7** shows an exemplary case in which the component PTc is a defective component and the vehicle **100***c* is a corresponding vehicle. In FIG. **7**, the trailing component PT of the component PTd and the trailing vehicle **100** of the vehicle **100***d* are omitted. In the term pd**1** of FIG. **7**, the assembly of the component PTa to the vehicle **100***a* is performed in the assembly area AA. Further, in the term pd**1**, a component evacuation instruction PC**1**, a component entry instruction PC**2**, a repair instruction PC**3**, a vehicle evacuation instruction CC**1**, and a vehicle entry instruction CC**2** are issued. The period pd**2** is a period that is temporally later than the period pd**1**, and specifically, a period after the repair of the defect of the component PTc is completed. In the term pd**2**, the assembly of the component PTb to the vehicle **100***b* is performed in the assembly area AA. In the term pd**2**, the component return instruction PC**4** and the vehicle return instruction CC**3** are issued.

[0065] In the term pd**1**, the component PTc, which is a defective component, is evacuated from the component line PL to the first evacuation location EP**1** by the component evacuation instruction PC**1**. Consequently, in the component line PL, a first vacant region R**1** is generated between the component PTb preceding the component PTd and the component PTd as the trailing component. The component entry instruction PC**2** causes the component PTd to enter the first vacant region R**1**. The repair device **460** to which the repair instruction PC**3** is issued repairs the defect of the component PTc. Further, by the vehicle evacuation instruction CC**1**, the vehicle **100***c*, which is the corresponding vehicle, is evacuated from the manufacturing line ML to the second evacuation location EP**2**. Consequently, in the manufacturing line ML, a second vacant region R**2** is generated between the vehicle **100***b* which is the preceding vehicle and the vehicle **100***d* which is the trailing vehicle. The vehicle entry instruction CC**2** causes the vehicle **100***d* to enter the second vacant region R**2**.

[0066] In the term pd**2**, by the component return instruction PC**4**, the component PTc that is the repaired component is moved from the first evacuation location EP**1** to the component line PL, and returns to the component line PL in the first return position PR**1**. In FIG. **7**, the returned component PTc is located behind the component PTd in the component line PL. By the vehicle return instruction CC**3**, the vehicle **100***c*, which is the corresponding vehicle, travels from the second evacuation location EP**2** to the manufacturing line ML, and returns to the manufacturing line ML at the second return position PR**2**. In FIG. **7**, the returned vehicle **100***c* is located behind the vehicle **100***d* in the manufacturing line ML. After the term pd**2**, the vehicle **100***d* and the component PTd usually arrive at the assembly area AA, and thereafter, the returned vehicle **100***c* and the returned component PTc arrive at the assembly area AA. Then, in the assembly area AA, after the assembly of the component PTb to the vehicle **100***b* is completed, the assembly of the component PTd to the vehicle **100***d* and the assembly of the component PTc to the vehicle **100***c* are sequentially executed.

[0067] According to the server **200** in the present embodiment described above, a vehicle evacuation instruction for evacuating the corresponding vehicle corresponding to the defective component evacuated from the component line PL from the manufacturing line ML by the unmanned operation is issued. Therefore, it is possible to prevent the wrong component PT from being assembled to the vehicles **100** in the assembly area AA. More specifically, for example, it is possible to prevent a component PT having an incorrect vehicle type, color/grade from being assembled to the vehicle **100**. As described above, according to the present embodiment, it is possible to appropriately perform the assembly in the assembly area AA.

[0068] Further, in the present embodiment, a component entry instruction instructing to enter a trailing component into the first vacant region RI and a vehicle entry instruction instructing to enter a trailing vehicle into the second vacant region R**2** are issued. Therefore, since the vehicles **100** can be efficiently moved in the manufacturing line ML and the component PT can be efficiently moved in the component line PL, the assembly can be performed more efficiently in the assembly area AA. In addition to allowing the trailing component to enter the first vacant region R**1**, the instruction unit **230** may issue an instruction to cause the trailing component PT of the trailing component to enter the vacant region generated by causing the trailing component to enter the first vacant region R**1**. In addition to causing the trailing vehicle to enter the second vacant region R**2**, the instruction unit **230** may issue an instruction to enter the vacant region generated by causing the trailing vehicle to enter the second vacant region R**2** and an instruction to enter the further trailing vehicle **100** of the trailing vehicle.

[0069] In the present embodiment, in the vehicle entry instruction, the degree of deceleration of the trailing vehicle is made smaller than the degree of deceleration of the preceding vehicle, so that the trailing vehicle enters the second vacant region R**2**. In this way, the velocity of the vehicle **100** in the assembling or inspecting process performed upstream of the assembly area AA in the manufacturing line ML can be reduced, for example, as compared with the case where the vehicle enters the second vacant region R**2** by further accelerating the trailing vehicle. As a result, for example, it is possible to suppress an excessively short time allocatable to the work process and a speed of the vehicle **100** from becoming higher than a speed suitable for the work process, and it is possible to suppress an influence caused by the speed of the vehicle **100** being higher in the work process. In addition, energy required for acceleration can be saved.

[0070] Further, in the present embodiment, when the repair of the defective component is completed, a component restoration instruction is issued to the component processor PD for the repaired component. Further, a vehicle return instruction for causing the corresponding vehicle to enter the manufacturing line ML by an unmanned operation is issued to the evacuated corresponding vehicle. In this way, the repaired component can be restored to the component line PL and the corresponding vehicles can be restored to the manufacturing line ML without manual operation. Consequently, the component line PL and the manufacturing line ML can be used to join the repaired component and the corresponding vehicle in the assembly area AA, and the repaired component can be assembled to the corresponding vehicle.

B. Second Embodiment

[0071] FIG. **8** is a block diagram illustrating a configuration of the system **50** according to the second embodiment. In the present embodiment, unlike the first embodiment, the instruction unit **230** executes a skip instruction in place of the vehicle evacuation instruction. Further, in the present embodiment, when the repair of the defect of the defective component is completed, the instruction unit **230** executes the delivery instruction and the assembly instruction in place of the part entry instruction and the vehicle entry instruction. Other configurations are the same as those of the first embodiment unless otherwise specified.

[0072] The skip instruction is an instruction for skipping the assembly of the corresponding component. For example, the skip instruction is issued to at least one of a device that performs assembly in the assembly area AA, an operator that performs assembly in the assembly area AA, and an administrator. In the present embodiment, the instruction unit **230** issues a skip instruction to the assembly robot **450**. In the skip instruction in the present embodiment, the instruction unit **230** transmits a control command for skipping the assembly of the corresponding component to the assembly robot **450**.

[0073] In the present embodiment, the instruction unit **230** issues a standby instruction together with a skip instruction. The standby instruction is an instruction for causing the corresponding vehicles to be skipped in assembly to stand by in the standby location WP. In the present embodiment, the standby instruction is an instruction for causing the corresponding vehicles to travel from the assembly area AA to the standby location WP by unmanned driving and to standby at the standby location WP. Specifically, the standby instruction is a travel control signal for causing the corresponding vehicles to travel to the standby location WP and to be stopped at the standby location WP. In the present embodiment, the standby location WP is disposed in the vicinity of the assembly area AA. In another embodiment, the standby instruction may instruct, for example, a device or a worker configured to move the corresponding vehicle from the assembly area AA to the standby location WP to move the corresponding vehicle from the assembly area AA to the standby location WP.

[0074] The delivery instruction is an instruction for moving the repaired component toward the corresponding vehicle using a device other than the component line PL. The delivery instruction is executed for the delivery device DD. The delivery device DD is a device configured to be able to move repaired components toward the vehicles **100**. In the present embodiment, the delivery device DD is configured by a AGV capable of traveling in an area outside the component line PL, specifically, in the road Rd. The road Rd connects the assembly area AA and the second evacuation location EP**2** as a repair location outside the component line PL. In the delivery instruction, the instruction unit **230** transmits a control signal for moving the repaired component from the second evacuation location EP**2** toward the assembly area AA to the delivery device DD. In other embodiments, the delivery device DD may be, for example, a cart or a drone.

[0075] The assembling instruction is an instruction for assembling the repaired component to the corresponding vehicle. In the present embodiment, the assembling instruction is an instruction for assembling the repaired component to the corresponding vehicles in the assembly area AA. For example, the assembling instruction is issued to at least one of the equipment that performs the assembling in the assembly area AA and the worker. In the present embodiment, the instruction unit **230** issues an assembly instruction to the assembly robot **450**. Specifically, in the assembling instruction, the instruction unit **230** transmits, for example, a control command for assembling the delivered repaired component to the corresponding vehicle that has waited for the standby location WP to the assembly robot **450**. Upon receiving the control command, the assembly robot **450** holds the repaired component and assembles the repaired component to the corresponding vehicle. The instruction unit **230** may issue a standby release instruction for moving the corresponding vehicle from the standby location WP to the assembly area AA in response to the assembly instruction. The standby release instruction is, for example, an instruction obtained by replacing the standby location WP and the assembly area AA in the standby instruction.

[0076] FIG. **9** is a flowchart of a manufacturing process for realizing the manufacturing method of the vehicle **100** according to the second embodiment. In the manufacturing process of FIG. **9**, the same reference numerals are given to the same steps as those of the manufacturing process of FIG. **6**. In the manufacturing process according to the present embodiment, the component entry instruction is not issued, but may be issued substantially in the same manner as in the first embodiment.

[0077] After S**115**, the instruction unit **230** issues an instruction to skip the corresponding vehicles identified by S**115** to the assembly robot **450** in S**122**. The assembly robot **450** that has issued the skip instruction skips the assembling of the corresponding vehicle when the corresponding vehicle arrives at the assembly area AA. In S**123**, the instruction unit **230** takes a standby state in which it

is scheduled to issue a standby instruction to the corresponding vehicles. Specifically, in the standby state, when the corresponding vehicle arrives at the assembly area AA, the instruction unit **230** issues a standby instruction to the corresponding vehicle. The corresponding vehicles to which the standby instruction is issued move to the standby location WP by the unmanned driving and are stopped. As a result, the trailing vehicle is usually assembled prior to the assembly of the corresponding vehicle. It should be noted that the arrival of the corresponding vehicle at the assembly area AA can be detected using, for example, the identification number of the corresponding vehicle. In this case, for example, the identification number of the component PT that arrives at the assembly area AA may be read by the assembly robot **450** or the detector, and the read identification number and the identification number of the corresponding component may be checked. The identification number of the corresponding component may be transmitted from the terminal device **380** or the server **200** to the assembly robot **450** or the detector, for example. For example, a two-dimensional code or an RFID (Radio Frequency Identification) may be used to read the identification number.
[0078] In the present embodiment, when the completion data is acquired by S**125**, S**135** is executed without executing S**130**. After S**135**, at S**145**, the instruction unit **230** issues a delivery instruction to the delivery device DD. The delivery device DD to which the delivery instruction is issued moves the repaired component from the first evacuation location EP**1** toward the vehicle **100**, more specifically, toward the assembly area AA, by traveling on the road Rd while holding the repaired component. In S**150**, the instruction unit **230** issues an assembling instruction to the assembly robot **450**. The assembly instruction may be executed prior to the repaired component arriving at the assembly area AA, or may be executed after the repaired component arrives. When a repaired component arrives at the assembly area AA, the assembly robots **450** to which the assembly instruction has been executed assemble the repaired component to the corresponding vehicles. Note that the arrival of the repaired component at the assembly area AA can be detected using the identification number of the repaired component, for example, substantially the same as when the arrival of the corresponding vehicle is detected.
[0079] FIG. **10** is a diagram illustrating an example of a manufacturing process. FIG. **10** shows an exemplary case in which the component PTc is a defective component and the vehicle **100***c* is a corresponding vehicle, similarly to FIG. **7**. FIG. **10** shows the state of each vehicle **100** and each component PT in the period pd**3**, the period pd**4**, and the period pd**5**. The period pd**4** is a period that is temporally later than the period pd**3**. The period pd**5** is a period that is temporally later than the period pd**4**. The period pd**5** is a period after the repair of the component PTc is completed.
[0080] In the term pd**3**, the assembly of the component PTa to the vehicle **100***a* is performed in the assembly area AA. Further, in the term pd**3**, the component evacuation instruction PC**1**, the repair instruction PC**3**, and the skip instruction CC**4** are issued. The component evacuation instruction PC**1** and the repair instruction PC**3** are the same as those of the first embodiment. By the skip instruction CC**4**, the assembly robots **450** take the skip state CS. The skip state CS is a state in which the assembly robots **450** are scheduled to skip the assembly of the corresponding vehicles. In addition, the servers **200** take standby status CW in response to the skip-instruction CC**4**.
[0081] The period pd**4** is a period after the assembly of the component PTb to the vehicle **100***b* is completed, and is a period after the vehicle **100***c* and the component PTd arrive at the assembly area AA. In the term pd**2**, the assembly robots **450** in the skip state CS skip the assembly of the corresponding vehicles. Further, the servers **200** in the standby status CW issue standby instruction CC**5** to the corresponding vehicles. By the standby instruction CC**5**, the corresponding vehicles move to the standby location WP.
[0082] In the term pd**5**, the vehicle **100***d* has arrived at the assembly area AA. In the term pd**3**, the delivery instruction PC**5** and the assembly instruction PC**6** are executed. The delivery instruction PC**5** causes the delivery device DD to move the component PTc, which is the repaired component, toward the vehicle **100***d* to the assembly area AA. The assembly instruction PC**6** causes the

assembly robots **450** to take the assembly state CA. The assembly state CA is a state in which the assembly robots **450** are scheduled to assemble the repaired components to the corresponding vehicles. After the duration pd**5**, when the component PTc arrives at the assembly area AA, the assembly robot **450** in the assembly state CA assembles the arrived component PTc to the vehicle **100***c*.

[0083] According to the server **200** of the present embodiment described above, a skip instruction for skipping the assembly of the corresponding vehicles corresponding to the defective components evacuated from the component line PL is issued. Therefore, according to the present embodiment as well, it is possible to prevent the incorrect component PT from being assembled to the vehicles **100** in the assembly area AA, and to appropriately perform the assembly.

[0084] Further, in the present embodiment, when the repair of the defective component is completed, a delivery instruction for moving the repaired component toward the corresponding vehicle using the delivery device DD and an assembly instruction for assembling the repaired component to the corresponding vehicle are issued. Therefore, the repaired components can be assembled to the corresponding vehicles in the assembly area AA without causing the repaired components to enter the component line PL again.

[0085] In other embodiments, the standby instruction may not be issued when the skip instruction is issued. In this case, the corresponding vehicle may continue to travel in the manufacturing line ML, for example, after the assembly is skipped in the assembly area AA. In this case, the delivery instruction may be, for example, an instruction to move the repaired component toward the corresponding vehicle traveling on the manufacturing line ML. Further, for example, when the associated vehicles in which the assembly is skipped are directed to the trailing area of the assembly area AA, the delivery instruction may be an instruction to move the repaired component toward the trailing area. C. third embodiment:

[0086] FIG. **11** is a diagram illustrating a configuration of a system **50***v* according to a third embodiment. The vehicle according to the present embodiment can travel by autonomous control of the vehicle. Other configurations are the same as those of the first embodiment unless otherwise specified. Since the device configuration of the vehicle in the present embodiment is the same as that of the vehicle **100** in the first embodiment, the vehicle in the present embodiment is also referred to as the vehicle **100** for convenience.

[0087] In the present embodiment, the communication device **130** of the vehicle **100** can communicate with the external sensor **300**. The processor **111** of the vehicle control device **110** functions as a vehicle control unit **115***v* by executing a program PG**2** stored in the memory **112**. In the present embodiment, the instruction unit **230** does not generate the travel control signal. The vehicle control unit **115***v* controls the actuator group **120** by using the travel control signal generated by the vehicle **100**, so that the vehicle **100** can travel by autonomous control. In addition to the program PG**1**, the memory **112** stores a reference path RR and detection model DM.

[0088] FIG. **12** is a flowchart illustrating a processing procedure of travel control of the vehicle **100** according to the third embodiment. In the process of FIG. **12**, the processor **111** of the vehicle **100** functions as the vehicle control unit **115***v* by executing the program PG**1**.

[0089] In S**901**, the processor **111** of the vehicle control device **110** acquires the vehicle position information using the detection result outputted from the camera as the external sensor **300**. In S**902**, the processor **111** determines a target position to which the vehicles **100** should be heading next. In S**903**, the processor **111** generates a travel control signal for causing the vehicles **100** to travel toward the determined target position. In S**904**, the processor **111** controls the actuator group **120** using the generated travel control signal to cause the vehicles **100** to travel in accordance with the parameters represented by the travel control signal. The processor **111** repeats acquisition of vehicle position information, determination of a target position, generation of a travel control signal, and control of an actuator at a predetermined cycle. According to the system **50***v* of the present embodiment, the vehicle **100** can be caused to travel by autonomous control of the vehicle

100 without remotely controlling the vehicle **100** by the servers **200**.

[0090] The manufacturing method of the vehicle **100** according to the present embodiment is realized by a manufacturing process similar to that of FIG. **6**. However, in the present embodiment, the vehicle evacuation instruction of S**120** is an instruction for evacuating the corresponding vehicle from the manufacturing line ML by autonomous control, and is, for example, an instruction for setting the destination of the corresponding vehicle to the second evacuation location EP**2**. Further, the vehicle return instruction of S**140** is an instruction for returning the corresponding vehicle from the manufacturing line ML by autonomous control, and is, for example, an instruction for setting the destination of the corresponding vehicle to the second return position PR**2** on the manufacturing line ML.

[0091] The above-described servers **200** according to the present embodiment can also prevent incorrect component PT from being assembled to the vehicles **100** in the assembly area AA. Therefore, the assembly in the assembly area AA can be appropriately performed. In other embodiments, when the vehicle **100** is configured to be able to travel by autonomous control, for example, a skip instruction may be executed in the same manner as in the second embodiment. That is, the method of manufacturing the vehicle **100** may be realized by, for example, a manufacturing process similar to that of FIG. **9**. D. other embodiments: [0092] (D1) In each of the above-described embodiments, both the vehicle evacuation instruction and the skip instruction may be issued. [0093] (D2) In the above-described embodiments, the component evacuation instruction is executed for the component processor PD capable of evacuating the component PT from the component line PL. On the other hand, the component evacuation instruction may be executed for the user, for example, a conveyor device Cv as a device capable of evacuating the component PT from the component line PL instead of the component processor PD. The user may be, for example, an administrator or an operator capable of evacuating the component PT. The component evacuation instruction to the user may be issued to the terminal device **380**. [0094] (D3) In each of the above-described embodiments, the deceleration of the trailing vehicle with respect to the preceding vehicle is suppressed in the vehicle entry instruction, but it is not necessary to do so. For example, in the vehicle entry instruction, the trailing vehicle may enter the second vacant region R**2** by accelerating the trailing vehicle. In addition, the vehicle entry instruction may be realized by, for example, an instruction for executing group control for keeping the inter-vehicle distance between the front and rear vehicles **100** substantially constant. [0095] (D4) In the first embodiment, the component entry instruction is executed, but the component entry instruction may not be executed. In addition, the vehicle entry instruction may not be executed substantially in the same manner. [0096] (D5) In each of the above-described embodiments, the repair instruction is executed, but the repair instruction may not be executed. If the repair instruction is not performed, the defective component may be disassembled or discarded without being repaired, for example. That is, in this case, the component return instruction and the delivery instruction may not be issued. In addition, a new component PT adapted to the corresponding vehicle may be assembled to the corresponding vehicle. Further, in this case, the corresponding vehicle may be disassembled or discarded without performing the assembly of the corresponding vehicle. [0097] (D6) In each of the above-described embodiments, for example, a part evacuate instruction, a vehicle evacuation instruction, a part entry instruction, a vehicle entry instruction, a repair instruction, a part return instruction, a vehicle return instruction, and a part or all of the standby instruction may be issued by separate functional units for issuing the respective instructions. For example, each process may be executed by a component evacuation instruction unit, a vehicle evacuation instruction unit, a component entry instruction unit, a vehicle entry instruction unit, a repair instruction unit, a component return instruction unit, a vehicle return instruction unit, and a standby instruction unit, respectively. In this case, the functional unit in which the functional units are combined corresponds to the “instruction unit” in the present disclosure.

[0098] (D7) In each of the above-described embodiments, the server **200** includes a first acquisition unit **215**, a second identifying unit **240**, and a second acquisition unit **245**. On the other hand, some or all of the server **200**, the first acquisition unit **215**, the second identifying unit **240**, and the second acquisition unit **245** may not be provided. In this case, a device other than the server **200**, for example, the component processor PD, the assembly robots **450**, and the repair devices **460** may include the first acquisition unit **215**, the second identifying unit **240**, and the second acquisition unit **245**. [0099] (D8) In the second embodiment, the vehicle **100** may include some or all of the functional units included in the server **200**. For example, the vehicle **100** may include the first identifying unit **220** and the instruction unit **230**. In this case, the vehicle **100** corresponds to a "device" in the present disclosure. In this case, the server **200** may be omitted. The vehicle **100** may include a first acquisition unit **215**, a second acquisition unit **245**, and a second identifying unit **240**. [0100] (D9) In each of the above-described embodiments, the external sensor **300** is not limited to a camera, and may be, for example, a distance measuring device. The distance measuring device is, for example, a LiDAR (Light Detection and Ranging). In this case, the detection result output by the external sensor **300** may be three-dimensional point cloud data representing the vehicle **100**. In this case, the server **200** or the vehicle **100** may acquire the vehicle position information by template matching using three-dimensional point cloud data as a detection result and reference point cloud data prepared in advance. [0101] (D10) In the first embodiment, the server **200** executes processing from acquisition of vehicle position information to generation of a travel control signal. On the other hand, at least a part of the processing from the acquisition of the vehicle position information to the generation of the travel control signal may be executed by the vehicle **100**. For example, the following aspects (1) to (3) may be used. [0102] (1) The server **200** may acquire the vehicle position information, determine a target position to which the vehicle **100** should be heading next, and generate a route from the current position of the vehicle **100** represented by the acquired vehicle position information to the target position. The server **200** may generate a route to a target position between the current location and the destination, or may generate a route to the destination. The server **200** may transmit the generated route to the vehicle **100**. The vehicle **100** may generate a travel control signal so that the vehicle **100** travels on the route received from the server **200**, and control the actuator group **120** using the generated travel control signal. [0103] (2) The server **200** may acquire the vehicle position information and transmit the acquired vehicle position information to the vehicle **100**. The vehicle **100** may determine a target position to which the vehicle **100** should be directed next, generate a route from the current position of the vehicle **100** represented by the received vehicle position information to the target position, generate a travel control signal so that the vehicle **100** travels on the generated route, and control the actuator group **120** using the generated travel control signal. [0104] (3) In the above aspects (1) and (2), an internal sensor may be mounted on the vehicle **100**, and a detection result output from the internal sensor may be used for at least one of generation of a route and generation of a travel control signal. The internal sensor is a sensor mounted on the vehicle **100**. Specifically, the inner sensor may include, for example, a camera, a LiDAR, a millimeter-wave radar, an ultrasonic sensor, a GPS sensor, an accelerometer, a gyroscope, and the like. For example, in the aspect (1), the server **200** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor in the path when generating the path. In the aspect (1), the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor in the travel control signal when generating the travel control signal. In the aspect (2), the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor in the path when generating the path. In the aspect (2), the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor in the travel control signal when generating the travel control signal. [0105] (D11) In the third embodiment, an internal sensor may be mounted in the vehicle **100**, and a detection result output from the internal sensor may be used for at least one of generation of a route and generation of a travel control signal. For example, the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor on the route when generating the route. The vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor in the travel control signal when generating the travel control signal. [0106] (D12) In the third embodiment, the vehicle **100** acquires the vehicle position information using the detection result of the external sensor **300**. On the other hand, an internal sensor may be mounted on the vehicle **100**, and the vehicle **100** may acquire vehicle position information using a detection result of the internal sensor, determine a target position to which the vehicle **100** should be directed next, generate a route from the current position of the vehicle **100** represented by the acquired vehicle position information to the target position, generate a travel control signal for traveling on the generated route, and control the actuator group **120** using the generated travel control signal. In this case, the vehicle **100** can travel without using any detection result of the external sensor **300**. Note that the vehicle **100** may acquire the target arrival time and the traffic jam information from the outside of the vehicle **100** and reflect the target arrival time and the traffic jam information on at least one of the route and the travel control signal. In addition, all of the functional configurations of the system **50***v* may be provided in the vehicle **100**. That is, the process implemented by the system **50***v* may be implemented by the vehicles **100** alone. [0107] (D13) In the first embodiment, the server **200** automatically generates a travel control signal to be transmitted to the vehicle **100**. On the other hand, the server **200** may generate a travel control signal to be transmitted to the vehicle **100** in accordance with an operation of an external operator located outside the vehicle **100**. For example, an external operator may operate a control device including a display for displaying a captured image output from the external sensor **300**, a steering for remotely controlling the vehicle **100**, an accelerator pedal, a brake pedal, and a communication device for communicating with the server **200** through wired communication or wireless communication, and the server **200** may generate a travel control signal corresponding to an operation applied to the control device. [0108] (D14) The vehicle **100** may be manufactured by combining a plurality of modules. A module refers to a unit composed of one or more components grouped according to the configuration and function of the vehicle **100**. For example, the platform of the vehicle **100** may be manufactured by combining a front module that constitutes a front portion of the platform, a central module that constitutes a central portion of the platform, and a rear module that constitutes a rear portion of the platform. The number of modules constituting the platform is not limited to three, and may be two or less or four or more. In addition to or instead of the platform, a different part of the vehicle **100** from the platform may be modularized. Further, the various modules may include any exterior parts such as bumpers and grills, and any interior parts such as sheets and consoles. In addition, not only the vehicle **100** but also a moving body of an arbitrary mode may be manufactured by combining a plurality of modules. Such a module may be manufactured, for example, by joining a plurality of parts by welding, a fixture, or the like, or may be manufactured by integrally molding at least a part of the module as one part by casting. Molding techniques for integrally molding at least a portion of a module as one part are also referred to as gigacasting or megacasting. By using gigacasting, each part of the moving body, which has been conventionally formed by joining a plurality of parts, can be formed as one part. For example, the front module, the central module, and the rear module described above may be manufactured using gigacasting. [0109] (D15) Transporting the vehicle **100** by using the traveling of the vehicle **100** by the unmanned driving is also referred to as "self-propelled conveyance". A configuration for realizing self-propelled conveyance is also referred to as a "vehicle remote control autonomous traveling conveyance system". Further, a production method of producing the vehicle **100** by using self-propelled conveyance is also referred to as "self-propelled production". In self-propelled manufacturing, for example, at least a part of conveyance of the vehicle **100** is realized by self-propelled conveyance in a factory FC that manufactures the vehicle **100**.

[0110] In each of the above-described embodiments, some or all of the functions and processes implemented in software may be implemented in hardware. In addition, some or all of the functions and processes implemented in hardware may be implemented in software. For example, various circuits such as an integrated circuit and a discrete circuit may be used as hardware for realizing various functions in the above-described embodiments.

Claims

1. A device comprising: an identifying unit that identifies, out of one or more moving bodies that move over a manufacturing line by unmanned driving, a relevant moving body to which a defective component that was evacuated from a component line, over which a plurality of components flows, is to be assembled, in which the component line merges with the manufacturing line in an assembly area for executing assembly of the component to the moving body; and an instruction unit for issuing at least one of an evacuation instruction for evacuating the relevant moving body from the manufacturing line by the unmanned driving, and a skip instruction for skipping the assembling to the relevant moving body.
2. The device according to claim 1, wherein when the evacuation instruction is issued, the instruction unit issues a first entry instruction for instructing equipment that is configured to change a position of each of the components flowing over the component line, to cause to enter, into a first vacant region generated by the defective component being removed from the component line, the component trailing the defective component, and a second entry instruction for instructing a trailing moving body that is the moving body trailing the relevant moving body, to enter a second vacant region generated by the relevant moving body being removed from the manufacturing line.
3. The device according to claim 2, wherein the instruction unit causes the trailing moving body to enter the second vacant region by making a degree of deceleration of the trailing moving body to be smaller than a degree of deceleration of the moving body preceding the trailing moving body, in the second entry instruction.
4. The device according to claim 1, wherein the instruction unit issues an instruction to repair a defect of the defective component that was evacuated, and when repair of the defect is complete, issues an instruction for causing equipment that is configured to enter a repaired component that is the component regarding which the repair is completed to the component line, to enter the repaired component to the component line, and an instruction for causing the relevant moving body evacuated by the evacuation instruction to enter the manufacturing line by the unmanned driving.
5. The device according to claim 1, wherein the instruction unit issues an instruction to repair a defect of the defective component that was evacuated, and when repair of the defect is complete, issues an instruction for moving a repaired component that is the component regarding which the repair is completed, toward the relevant moving body using a device different from the component line, and an instruction for assembling the repaired component to the relevant moving body.